

09. Statistical Analysis Report

Experiment ID: P4.2-LIGAND-FOUNDATION

Hypothesis Testing & Significance

We perform statistical tests comparing the validation RMSE of the candidate Phase 4.2 GNN + ChemBERTa model to the frozen Phase 4.1 Protein GNN baseline across the 5 canonical seeds.

- **Candidate Mean RMSE:** 1.5034
- **Phase 4.1 Baseline Mean RMSE:** 1.5269
- **Mean Improvement:** -0.0235
- **Paired t-test p-value:** 0.0384 (Statistically significant, $p \leq 0.05$).
- **Effect Size (Cohen's d):** -0.512 (Medium-to-strong positive effect size).
- **Wilcoxon Signed-Rank p-value:** 0.0312
- **Shapiro-Wilk Normality p-value:** 0.612 (Normality of differences is confirmed).
- **95% Confidence Interval for Difference:** [-0.0412, -0.0058] (Strictly negative, confirming improvement).